

Nathaniel Viana

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Education

Rensselaer Polytechnic Institute

Bachelors of Science in Computer Science
GPA: 3.36/4.0

Troy, NY
Expected May 2027

Relevant Coursework

Data Structures, Introduction to Algorithms, Introduction to AI, Computer Organization

Experience

FSS Intern

GlobalFoundries

Malta, NY
Mar 2026 - Present

- Designed and developed key features for an internal Python automation tool, accelerating factory systems verification processes and reducing cycle times.
- Mentored and onboarded team members on codebase architecture, system workflows, and technical documentation to ensure long-term maintainability.
- Collaborated with cross-functional users to gather requirements, translating user feedback into actionable, customized application features.

Projects

Vessel - Workout Tracking Application

Feb 2026 - Present

- Developed a RESTful backend with Spring Data JPA using logical layers and entity relationships.
- Applied architectural practices such as separation of concerns, and DTOs for secure request/responses.
- Containerized local development and PostgreSQL instances using Docker Compose for easy setup and reproducibility.

PSoft Tools - Open-Source Software Development Project

January 2025 - May 2025

- Contributed to the development of a software platform that helps students learn key computer science principles.
- Implemented tools for testing Hoare Triples and validating counterexamples using Java enhancing the applications usability.
- Collaborated with a team in an open-source environment, honing skills in React, Typescript, and Dafny.

RL Racetrack

April 2025

- Built a reinforcement learning agent using Q-learning to navigate a racetrack from start to finish without going off track.
- Integrated a pre-built Gymnasium-compatible environment to train a Q-learning agent, using Pygame for visualization.
- Tuned parameters and reward strategies to improve the convergence and learning rate of the agent.

Activities

Coding&&Community Vice President

September 2023 - May 2025

- Coordinated and executed regional community events, including hackathons and weekly workshops, to introduce foundational computer science concepts to young students.
- Developed introductory programming curriculums and directly instructed classes, teaching K-12 students the basics of software development.
- Managed a team of student instructors, oversaw event logistics, and collaborated with executive board members to expand club outreach.

Skills & Interests

Language Skills: English, Spanish

Technical Skills: Java, Python, Spring Boot, Docker, Git, PostgreSQL